

XIAOGE ZHANG, PhD

Cambridge, UK | **Full Right to Work in the UK (No sponsorship required)**

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Professional Profile

Health economist and quantitative modeller (PhD, York). Hands-on experience localising global cost-effectiveness and budget-impact models for Gulf HTA submissions, adapting the accompanying technical reports, completing HTA submission forms, and running independent QC on third-party models. Underpinned by 7+ years of academic research in econometrics, causal inference and applied statistical modelling. Works in R, Stan and Excel, with an interest in reproducible, auditable model quality assurance.

Technical Competencies

- **Economic Evaluation:** Cohort Markov and partitioned-survival models; cycle structure and half-cycle correction, background mortality, treatment waning, stopping rules; OWSA, scenario analysis, PSA, threshold analysis.
- **Budget Impact & Burden of Illness:** Epidemiological funnel, market share and uptake, expenditure caps and managed-entry arrangements; prevalence-based, bottom-up cost-of-illness design.
- **Evidence Synthesis & RWE:** Bayesian NMA, population-adjusted indirect comparison, ML-NMR; parametric survival extrapolation (Weibull, Gompertz, splines) and Kaplan–Meier data reconstruction; causal inference, propensity-based weighting (IPTW, AIPW) and external control arms; awareness of UK RWD sources (SACT, CPRD) in NICE submissions.
- **Utilities & Costing:** EQ-5D-5L/3L mapping to national value sets; NICE DSU TSD23 severity modifiers and QALY shortfall; micro-costing from national tariff schedules and published drug price lists.
- **Model Quality Assurance:** Formula pattern census, cross-sheet dependency mapping, dead-parameter and hardcode detection, whole-model diffing, reproducible scenario execution with verified source integrity, provenance grading of inputs.
- **Toolkit:** **Excel** (model building and debugging, Solver calibration, PSA/DSA dashboards); **R** (flexsurv, survival, meta, netmeta, tidyverse, Quarto reporting); **Stan** (hierarchical Bayesian models, MCMC diagnostics).

Professional Experience

HEOR Modeller

Masdar Al Hekmah (HEOR & market access consultancy)

Jun – Aug 2026

Riyadh, KSA — remote

- **Model Localisation:** Adapted global cost-effectiveness and budget-impact models for Saudi MOH and UAE DoH submissions across nine therapeutic areas: comparator and population configuration, utility mapping to local value sets, national tariff and drug-price sourcing, OWSA, scenario and PSA.
- **Model & Instrument Design:** Designed a prevalence-based, bottom-up burden-of-illness model for a new indication—architecture, module logic and input specification across direct, indirect and HRQoL components—with build documentation and a client-facing concept approved before implementation; separately designed a clinician discussion guide, audited for coverage against the model's input requirements.
- **Technical Reporting & Submissions:** Adapted technical reports (70–100pp) through a staged pipeline—table disposition, section tiering, anchored text change plans, numerical reconciliation against the model—and completed HTA submission forms for payer review.
- **Quality Assurance & Methods Tooling:** Ran independent QC on third-party models, identifying

unstable calculation states, unreferenced parameters and formula errors reaching published outputs; built reusable instruments including an EQ-5D-5L/3L mapping tool for two national value sets and a NICE DSU TSD23 QALY-shortfall template.

Lecturer → Associate Professor (Quantitative Economics & Modelling)

2018 – 2025

Southwestern University of Finance and Economics

Sichuan, China

- **Scientific Writing & Research Delivery:** Principal writer for a sub-project within a National Social Science Fund Major Grant (2020–2023); produced structured technical reports synthesising quantitative evidence under competitive peer review, and taught quantitative methods bilingually—translating complex modelling concepts for non-specialist audiences.
- **Quantitative Project Supervision:** Supervised 15 Master’s students through full research cycles—study design, analytical methodology, final reporting—managing concurrent projects with structured feedback and quality review at each stage.

Technical Portfolio

Methods work reconstructing published NICE appraisals. Code, models and write-ups at xgzhang.com.

Evidence Synthesis & External Control Arms (R/Stan)

2025–2026

- Reconstructed evidence networks and external control arms for single-arm oncology studies lacking direct comparators (e.g. TA1013, TA1017); applied population-adjusted indirect comparison and propensity-based weighting, with Bayesian hierarchical models in Stan and full MCMC diagnostics.
- Authored an eight-part NMA technical guide covering Bayesian NMA, NMR, survival NMA, CNMA, IPD-NMA, MAIC, STC and ML-NMR, with assumptions, implementation logic and HTA relevance. [Weblink](#)

Oncology Model Validation & Remediation (Excel/R)

2026

- Validated and remediated a NICE-style three-state partitioned survival model: resolved time-unit mismatches causing survival crossovers and implemented hazard chaining in R to remove discontinuities at the trial-to-extrapolation boundary; delivered as a reproducible workflow with an automated PSA/DSA dashboard.

NICE Appraisal Methodology Repository

2025–Present

- Reviewed 20+ NICE appraisals across oncology and rare disease, examining how methodological choices drive cost-effectiveness conclusions; distilled into 50+ technical reference cards covering ESS shrinkage, treatment waning and non-linear utility mapping.

Education

University of York, Ph.D. in Economics

York, UK | 2013 – 2018

Structural econometrics, causal inference, and simulation-based estimation methods.

University of Sheffield, M.Sc. in Business Finance and Economics

Sheffield, UK | 2011 – 2012

Selected Publications

Indirect inference for bias calibration in irregular-interval panel data models. *Economics Letters* (2023).

Firm risk-taking under monetary policy uncertainty. *Pacific-Basin Finance Journal* (2022).

Languages: English (full professional proficiency); Mandarin (native). | **Certification:** Associate Fellow (AFHEA), Advance HE, 2025.